

Database (Databaza)

Exam Task — Java OOP

Overview

The goal is to create a simple program for processing tabular data from text files. The program supports three operations: loading a table from a file, filtering rows by a condition, and selecting specific columns. Operations never modify the original table — each returns a new table. Operations can be nested (output of one is input of another).

Input File Format

Each file represents one table:

- Line 1: two integers — number of columns and number of data rows
- Line 2: column names (space-separated, lowercase, no spaces/diacritics)
- Remaining lines: data rows (space-separated values, no spaces in values)

Commands

```
subor <filename>                - load table from file
filter <col> <value> <operation> - filter rows where col==value
vyber <count> <col1> <col2> ... <operation> - select specific columns
```

Commands can be nested — the last token of filter/vyber is itself a command:

```
vyber 2 meno priezvisko filter rocnik 2 subor studenti.txt
```

Output Format

Each column is right-aligned in 10 characters, separated by |:

meno	priezvisko	bydlisko	rocnik	program
Martin	Kovac	Zilina	2	I

Class Descriptions

Tabulka (11 pts) — represents a table with columns and rows

- new(stlpce: String[]) — creates empty table with given columns
- vytvorPrazdnuKopiu(): Tabulka — creates new empty table with same columns
- pridajRiadok(hodnoty: String[]) (2 pts) — adds row; throws exception if wrong column count
- getIndexStlpca(stlpec: String) (2 pts) — returns column index by name; throws exception if not found
- getRiadky(): Iterable<Riadok> (1 pt) — returns all rows
- vypis() (3 pts) — prints table in output format

Riadok (6 pts) — stores values of one record

- new(tabulka: Tabulka, hodnoty: String[]) — creates row belonging to a table
- getHodnota(stlpec: String) (2 pts) — returns value for given column name
- kopirujDo(tabulka: Tabulka) (2 pts) — copies this row into another table

Operacia (2 pts) — interface for all operations

- vykonaj(): Tabulka

Subor (7 pts) — loads table from file

- new(nazovSuboru: String)
- vykonaj() (5 pts) — reads file, creates and returns table

Filter (5 pts) — filters rows by column value

- new(vstup: Operacia, stlpec: String, hodnota: String)
- vykonaj() (3 pts) — returns table with only rows where col==value

Vyber (5 pts) — selects specific columns

- new(vstup: Operacia, stlpce: String[])
- vykonaj() (3 pts) — returns table with only selected columns

Databaza (10 pts) — executes commands

- new() — creates database object
- vykonaj(prikaz: String) (2 pts) — parses and executes command, returns result table
- nacitaj(prikaz: String) (6 pts) — parses text command into Operacia object recursively

Main (4 pts) — loop: read command, if 'koniec' exit, else execute and print result**Points Summary**

Class	Points
Tabulka	11
Riadok	6
Operacia (interface)	2
Subor	7
Filter	5
Vyber	5
Databaza	10
Main	4
TOTAL	50